Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Smart Attend: AI-Powered Classroom Attendance and Monitoring System with Real-Time Location Verification

Rakshit - 2210990712
Archi Gandhi - 2210990150
Samreen Kaur - 2210992240
Navleen Kaur - 2210991965

Supervised By
Gurpreet Singh
Mentor

Department of Computer Science and Engineering,
Chitkara University, Punjab

Introduction

Background

Modern educational institutions face significant challenges with traditional attendance systems. Manual roll calls and sign-in sheets are time-consuming, error-prone, and vulnerable to proxy attendance.

Problem Statement

Existing attendance mechanisms fail to guarantee authenticity, waste instructional time, and lack integration with academic reporting systems. An intelligent, tamper-proof solution is required.

Solution Overview

SmartAttend is an AI-powered system that leverages facial recognition, live location verification, a static AI chatbot, and automated report generation to eliminate manual effort entirely.

Abstract

SmartAttend utilizes computer vision and facial recognition technology to automatically detect and identify students through a classroom-installed camera. The system processes real-time video streams to mark attendance accurately without manual intervention. It is designed to function effectively under varying lighting conditions, facial orientations, and partial occlusions.

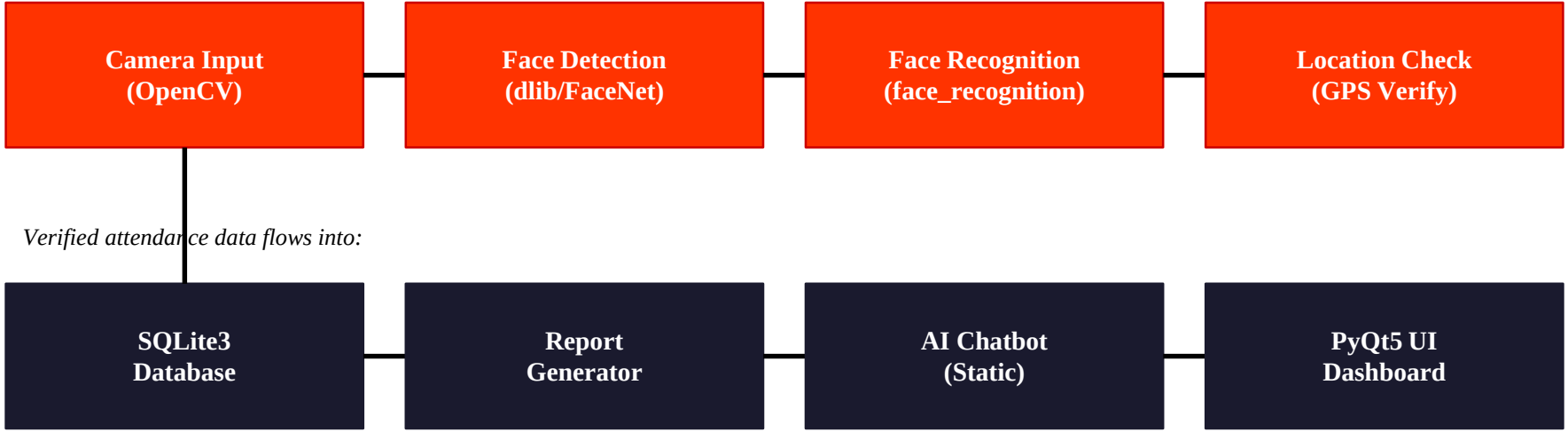
In addition to automated attendance, the system integrates live location detection to verify classroom presence. An embedded static AI chatbot assists users by providing attendance details and basic query handling. The platform generates structured attendance reports, enabling faculty and administrators to monitor student participation efficiently.

By combining facial recognition, location verification, AI-based interaction, and automated reporting, SmartAttend improves accuracy, eliminates proxy attendance, saves instructional time, and advances intelligent education management.

Objectives

- To develop an automated, real-time facial recognition system for student attendance marking.
- To integrate live GPS-based location verification to ensure classroom presence and prevent proxy attendance.
- To build a PyQt5-based desktop interface for seamless interaction by faculty and students.
- To embed a static AI chatbot capable of answering attendance queries and providing system information.
- To generate structured, downloadable attendance reports for administrative transparency.
- To ensure system reliability under varying lighting conditions, facial orientations, and partial occlusions.

System Architecture



Flow: Camera captures frames -> Face detected -> Face matched against DB -> Location confirmed -> Attendance marked -> Reports generated

Technology Stack

Frontend & Backend

- Language: Python 3.x
- UI Framework: PyQt5
- Architecture: Component-Based (reusable widgets, forms, dashboards)
- API Design: Python/PyQt5 for client-server data transfer & face recognition backend

AI / Database / Hardware

- Face Detection: OpenCV
- Face Recognition: dlib / FaceNet / face_recognition library
- Database: SQLite3 (persistent storage, efficient retrieval)
- Location: Live GPS/IP-based verification module
- Hardware: Classroom webcam / IP camera

Key Features

Facial Recognition

Real-time student identification via OpenCV + face_recognition; handles varying lighting & orientations.

Location Verification

Live GPS/IP-based classroom presence check to prevent proxy attendance.

AI Chatbot

Static embedded chatbot for attendance queries, system info, and user assistance.

Automated Reports

Structured attendance reports exportable for faculty and administrators.

PyQt5 Dashboard

Interactive desktop UI with reusable component-based architecture.

SQLite3 Storage

Lightweight persistent database for efficient attendance data management.

Software & Hardware Requirements

Software Requirements

- Python 3.x + PyQt5 (Frontend & Backend)
- OpenCV - real-time face detection from camera stream
- dlib / FaceNet / face_recognition - accurate facial recognition
- SQLite3 - lightweight database for persistent attendance storage
- GPS / IP Location Module - for classroom presence verification

Hardware Requirements

- Classroom camera (webcam or IP camera) for video stream input
- Computer/server with sufficient processing power for real-time CV tasks
- Network/Internet connection for GPS-based location verification
- Student devices (optional) for chatbot query interface access

Working Methodology

1

Camera Initialization

OpenCV captures live classroom video via installed camera.

2

Face Detection

Haar Cascades / dlib detects faces in each video frame.

3

Face Recognition

face_recognition library matches faces against registered student database.

4

Location Verification

GPS module verifies the student's physical classroom presence.

5

Attendance Marking

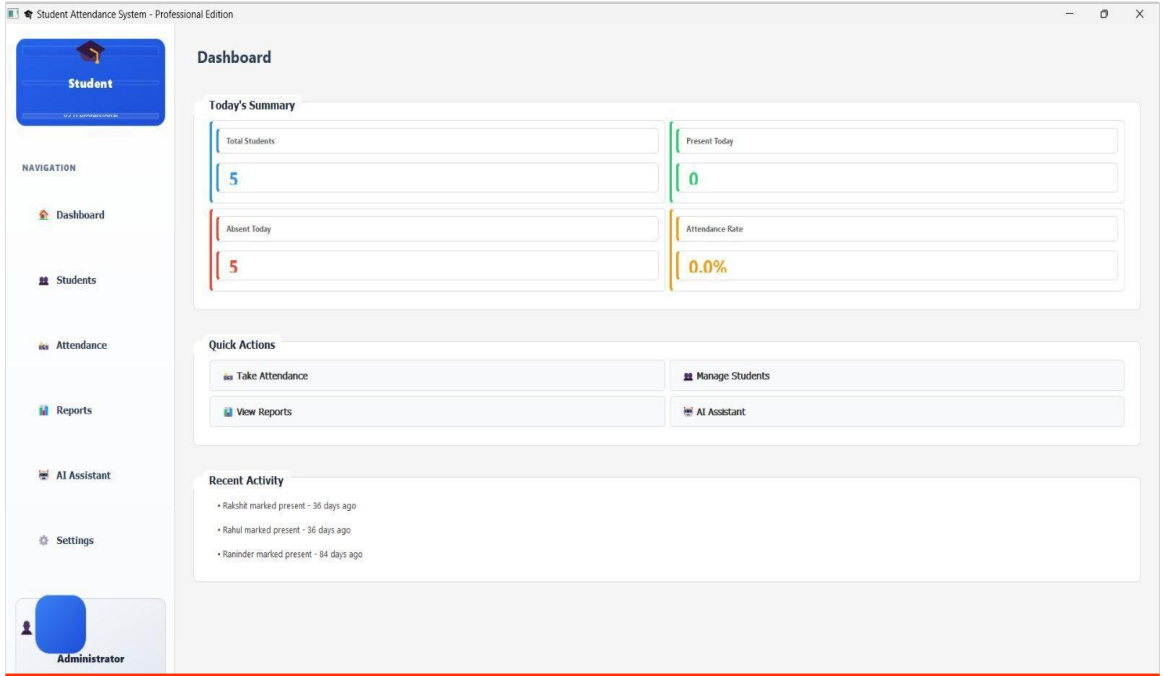
Verified students are marked present in SQLite3 database automatically.

6

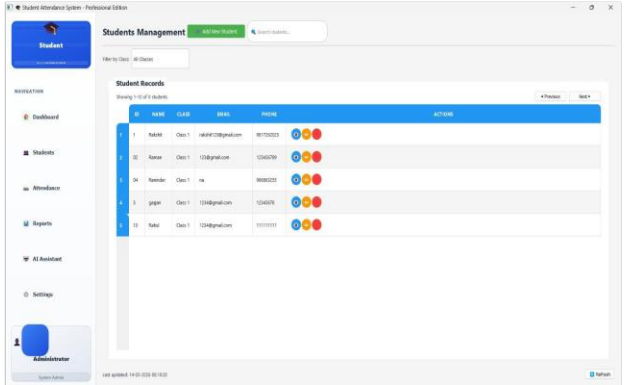
Report Generation

Structured reports generated and displayed via PyQt5 dashboard.

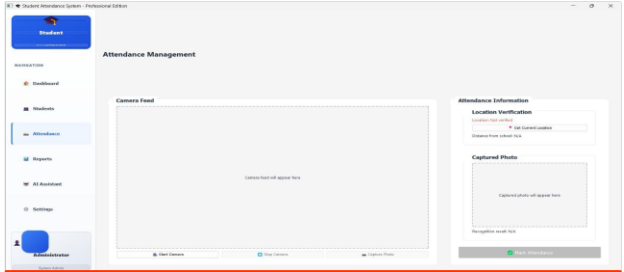
System Screenshots – UI Overview



Dashboard – Today's Summary & Quick Actions

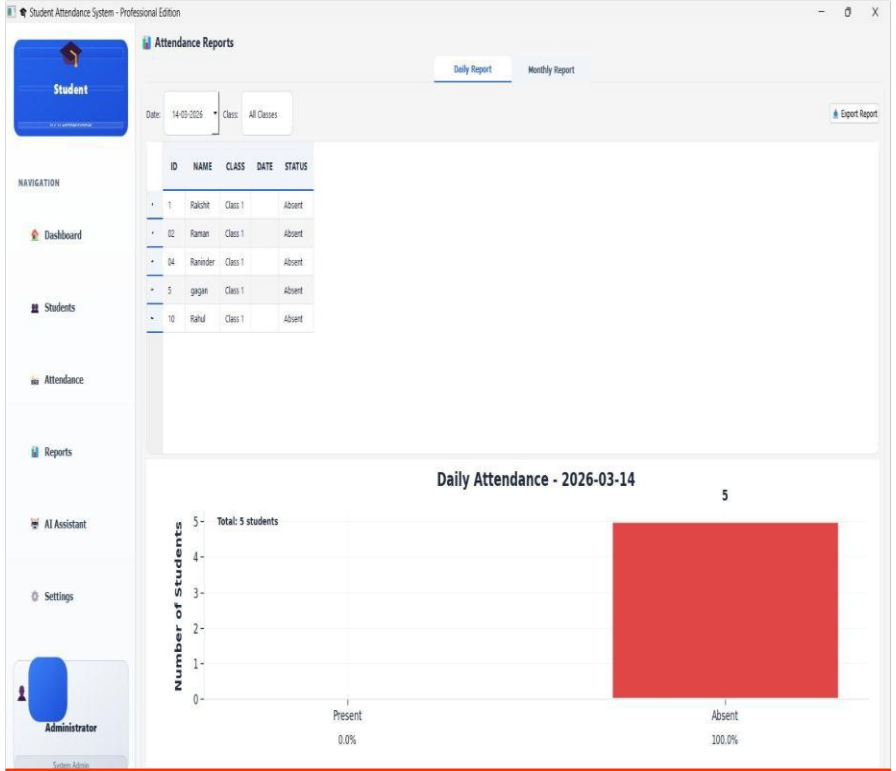


Student Records Management

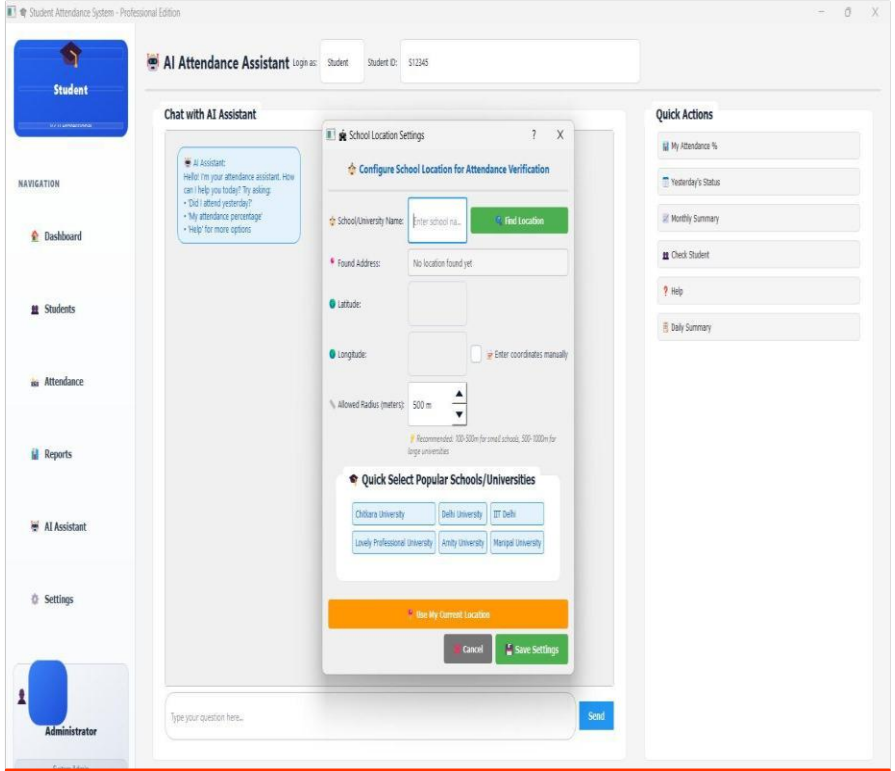


Attendance – Camera & Location Module

System Screenshots – Reports & AI Assistant



Attendance Reports – Daily View & Chart



AI Attendance Assistant – Chat Interface

Expected Outcomes

- Complete elimination of proxy attendance through dual-layer verification (face + location).
- Significant reduction in time spent on manual attendance, freeing up instructional time per session.
- High recognition accuracy (>95%) even under low-light conditions and partial facial occlusions.
- Fully automated, downloadable attendance reports accessible to faculty and administrators.
- Improved student accountability and transparent academic performance monitoring.
- A scalable system adaptable for multiple classrooms and departments across the university.

Conclusion

SmartAttend represents a significant step forward in the digitization of educational administration. By integrating computer vision, facial recognition, real-time location verification, and AI-driven interaction, the system delivers a robust and reliable attendance solution for modern classrooms.

The use of Python, PyQt5, OpenCV, face_recognition, and SQLite3 ensures a lightweight yet powerful architecture deployable with minimal infrastructure. The embedded chatbot further enhances user experience by providing instant answers to attendance-related queries.

Ultimately, SmartAttend contributes to academic integrity, operational efficiency, and the advancement of intelligent education management, aligning with the vision of smart campuses in the digital era.

Thank You